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Reflection Stamps: A simple and positive method to support self-evaluation in illustration drawing

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ABSTRACT

In this research, which targets illustration drawing, we designed, implemented and evaluated a system, called ‘Reflection Stamps’, to reduce the effort and mental burden of the act of reflecting on one’s work and encourage positive reflection. When drawing an illustration, it is important for the learner to reflect positively on their own work to aid the development of their technique. However, the majority of existing reflection methods require a great deal of effort, or else involve a negative approach of indicating points for improvement, which results in the problem that the act of reflection imposes a mental burden. The proposed ‘Reflection Stamps’ system is equipped with a word stamp function that makes it easy to record evaluation and feedback on each part of an illustration. The system is furnished with a large number of stamps with positive content to encourage the learner’s sense of self-affirmation. The results of the evaluation experiment showed that the proposed system reduced the time required for reflection by 60% in comparison to the comparative method, which was a conventional method. In addition, there was a 133% increase in positive reflections when compared to the comparative method, demonstrating that the proposed system facilitated many more positive reflections. Furthermore, the system received a rating of 79 points on the System Usability Scale, confirming that it has good usability. However, it also became clear that to increase operability it is necessary to improve the placement of the stamps, and the enlarge/reduce function. In future we aim to construct a system to better support skill improvement in illustration, by increasing the variety and range of expression of the stamps to enable effective and accurate reflection.

1. Introduction

Currently, regardless of whether they are professionals or amateurs, the number of people who draw illustrations for work or as a hobby is increasing. On the illustration posting platform ‘pixiv Inc. [1]’, the total number of registered users surpassed 84,000,000 in 2022, and the number of posted works exceeds 100,000,000 . Most people who are learning to draw teach themselves. In a survey of illustrators conducted by MUGENUPInc. [2], it was revealed that out of 2661 illustrators, including both those whose main job was illustration and those for whom it was a side business, 53% were self-taught. From this it can be concluded that supporting the development of drawing skills is important for people who draw illustrations. In addition, the process of reflecting on one’s activity and checking which parts have been done well and which parts need to be improved is important for skill development.

Resnick [3] proposed the creative thinking cultivation model, in which repeating the five component activities – Imagine, Create, Play, Share, Reflect – develops creative thinking. It is said that in the ‘Reflect’ stage looking back on the creative process enables one to understand and clearly express the thinking that forms the core of the creative approach. Thus, we consider reflection to be a vital factor in learning creative thinking.

Ishihara and Taizan [4] demonstrated that recording detailed reflections on cognitive activities significantly increases cognitive desire (a spontaneous trend in which people enjoy engaging in cognitive activities that require effort). It was proven that, in middle school classes, the cognitive desire of students who carried out detailed reflection on cognitive activities was significantly higher than that of students who did not carry out reflection. Furthermore, as a result of analyzing the content of the reflections, it was found that there was a significant increase in the cognitive desire of students who were able to record their reflections in detail, whereas a significant increase was not observed in students who recorded more abstract reflections. There is also research showing that reflection has the function of motivating the learner and reducing the sense of burden regarding learning, by making the learner feel enjoyment and interest at times when they are

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beginning to lose their motivation to learn. Accordingly, we consider the act of reflection to be important in skill improvement in illustration drawing as well.

In addition, it is known that positive reflection, i.e. looking back on one's activity and thinking 'I did it!', increases a learner's sense of self-efficacy. Self-efficacy refers to what is thought to be at work in the selection, undertaking, continuation, and completion of an activity. Learners who are highly self-efficient find it easier to participate in learning activities and to keep striving, even when faced with problems. Bandura et al. state that reflecting on one's experience of success produces a sense of self-efficacy, confidence, and desire toward a task Bandura [5]. When there is a positive sense of self-satisfaction in the reflection stage of the creative process, it is thought that motivation to learn increases in the form of a stronger awareness of self-efficacy regarding skill acquirement, a strong sense of the aim of learning, and a strong spontaneous interest in the task. Furthermore, according to Ellis et al. [6], it has been demonstrated that reflecting after one's experiences increases performance, regardless of whether those experiences are of success or of failure. Therefore, we consider that carrying out positive reflection will increase sense of self-efficacy and have a positive effect on the continuation and completion of learning.

Additionally, Baumer et al. [7] pointed out the ambiguity in the definition of reflection within interactive system design, emphasizing the importance of intentionally supporting reflection from the design stage. Sharmin and Bailey [8] empirically demonstrated the critical role reflection plays in creative design processes, providing guidelines for designing reflection-support tools. Furthermore, Kocielnik et al. [9] showed that conversational reflection-support systems effectively enhance users' motivation and self-efficacy, providing concrete evidence for the efficacy of technological support in promoting reflection.

As a means of evaluating an illustration, writing directly onto the illustration is becoming popular. On X (Twitter), there is a culture of posting an illustration one has drawn, with the hashtag '# I want you to write on my picture and praise me,' to invite praise from followers(Fig. 1). In addition, on 'sessa' by FlatplanInc. [10], a service in which professional creators can be asked to assess illustrations, it is common for the creators to suggest corrections by writing on the illustrations(Fig. 2). This evaluation method of writing can involve both other people writing on one's illustration, and writing on one's illustration by oneself.

However, the details of the methods of reflection are still unclear. It is not known how many people who have drawn illustrations are reflecting on those illustrations, nor whether there is a larger number of positive reflections or negative reflections. Therefore, in this research, as a prior survey, we first conducted a fact-finding survey relating to reflective behavior in illustration drawing.

We investigated whether learners were reflecting on their illustrations, and what methods of reflection they used. Using an online questionnaire, we surveyed 101 males and females who draw illustrations on a regular basis. The breakdown of the number of years for which the respondents had been drawing was as follows: less than 1 year: 12%, 1–3 years: 25%, 4–10 years: 20%, more than 10 years: 43%. In addition, 25% of the respondents had specialized training in fine art or illustration.

The survey results are as follows. In response to the question asking whether, after drawing a picture, they reflected on which parts were done well and which parts needed improving, 75% of respondents answered 'No'. Reasons for this included "Because it is tiresome", "Because I don't have time", and "Because I don't want to look at the pictures I draw (because I'm embarrassed about my poor drawing skills)". Regarding the content of reflections, it was found that 94% of respondents identified negative aspects of their drawings, such as parts that had not been drawn well. To give specific examples, comments included "I should have chosen a pose that looked cuter/cooler (more attractive)", "The face doesn't look like the person I was trying to draw", and "There's no sense of texture". It should be mentioned that

Fig. 1. Example of evaluative comments written on an illustration on X.

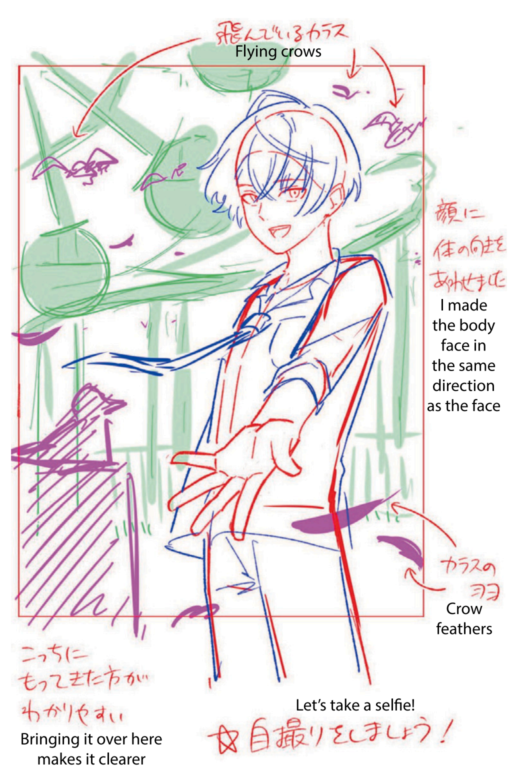

Fig. 2. Example of evaluative comments written on an illustration on the illustration correction website 'sessa'.

for this question respondents who did not usually reflect on their work were asked to imagine what reflections they would make.

The fact that the reasons given for not carrying out reflection included the time and effort required suggests that the perception of reflection as being hard work and tiring is a problem. Furthermore, it seems that most people have a negative impression of their own illustrations, and this perception is a factor that increases the mental burden imposed by reflection. The tendency of learners to focus only on negative aspects, such as what they failed at and what they think is bad, when reflecting on drawing, has also been pointed out by Kajiura and Kobayashi [11].

Therefore, in this research, we aim to design, implement, and evaluate a system, called 'Reflection Stamps' (hereafter referred to as the proposed system), that reduces the effort and mental burden of the act of reflecting on one's illustrations and promotes positive reflection. We propose a reflection method using word stamps, to make reflection simple. The stamps we create have content that is intended to encourage more positive reflections. We have learners use the proposed system, and verify whether the effort and mental burden of reflection is reduced. Whether or not reflection requires effort is evaluated based on the time taken for reflection and the rating on the System Usability Scale by Brooke [12], which is an index that measures how easy a system is to use. In the index of 'whether reflection can be carried out easily', we consider the conditions necessary to make it possible to use the system smoothly. Accordingly, we also evaluate system usability.

The positive reflection referred to in this research means praising oneself on a picture that one has drawn. In contrast, negative reflection means finding faults with one's drawing and working out how to address those faults.

2. Related works

As research into supporting reflection on illustration, there is the tool developed by Misue and Kowata [13]., which makes viewing 'making-of' teaching materials more efficient. 'Making-of' teaching materials refer to materials that present the process of how something is made and can be perceived as spatio-temporal data. The special features of the tool are that it can provide an overhead view of the events that occur during the construction process, and can excise spatio-temporal data within the campus space and focus on the events contained therein. However, the tool does not support specific reflection that involves identifying good points and points for improvement.

There is also research that applies the creative thinking cultivation model proposed by Resnick [3]. Morimoto and Takada [14] proposed a 'Spiral and Tag Model' for GUI programming, which applies the creative thinking cultivation model, and constructed a system to support the model. The 'Spiral and Tag Model' is a model in which sharing and reflecting, which were originally carried out after a creative activity, are carried out during the creative activity. To support this model, Morimoto et al. constructed a sharing system that enables users to share sketches and scripts and applied the system to a workshop that uses a GUI programming environment. As a result, it was found that subjects' reactions toward sharing and reflection differed depending on the method of sending and displaying a creative work. Nevertheless, this is a support system targeted at GUI programming education; there is no system that supports the reflection stage of the creative thinking cultivation model in illustration drawing.

Sugano and Nakamura [15] proposed a method of dividing an image into different viewpoints and verbalizing those viewpoints, to assist illustration beginners. When a beginner copies a face, there is the problem that while verbalizing the features of the face that is being copied increases the amount of detail in the drawing, the layout and size of the parts tend to diverge from the original. In response to this, Sugano et al. proposed a method that divides an image into separate viewpoints and verbalizes those viewpoints, and verified the resulting depth and multifaceted quality of observation in copying. As a result, their proposed method was shown to make verbalization multifaceted and specific, and improve observation regarding the target image as a whole. However, the research makes no mention of reflecting on drawing.

In the field of illustration-specific reflection support systems, no method has been proposed to carry out verbalization and reflection both simply and effectively. In our research, we compensate for this by proposing a new approach that simply supports specific reflection in illustration.

3. Design and implementation of reflection stamps

Reflection Stamps is a system in which reflection is carried out by placing stamps that show words of evaluation on an illustration. The reflection method involves creating specialized stamps for each part of the figure in the picture, such as the hair or face, and placing these stamps on arbitrary positions on an illustration.

3.1. Definition of requirements

The requirements of the proposed system are as follows:

- Easiness: Being able to carry out reflections easily and quickly. In particular, using text stamps makes it possible to carry out reflection with minimal effort, with no need to write detailed sentences.
- Encouraging positive reflection: Word stamps with positive content that indicate the parts of an illustration that are done well enable learners to notice anew the good points of their illustrations. This reduces a learner's negative impression of an illustration they have drawn and decreases the learner's mental burden.

Conventional reflection methods require the writing of detailed sentences, which makes reflection complicated.

In this research, we aim to realize easier reflection, by using word stamps.

In general, the more types of word stamp there are, the more effort will be required to search for a word stamp, which will reduce the simplicity of the method. On the other hand, if the types of word stamp are limited or the content of the available stamps is not relevant to reflection, suitable reflection becomes difficult. Therefore, in this research, which uses a wide variety of stamps, we conduct a prior experiment to investigate the word stamps utilized for reflection, with the aim of realizing both accuracy and simplicity.

3.2. Functions of the proposed system

The proposed system works by learners uploading illustrations they have drawn and placing various types of stamp on the illustrations to carry out reflection by themselves. The screen of the proposed system is presented in Fig. 3. A word stamp is selected, by clicking or tapping on one of the stamps shown in the stamp list, and placed in the illustration area. Stamps placed in the illustration area can easily be moved by dragging and dropping. Thus, it is possible visually to record components that demonstrate one's successes as well as points that need improving. Stamps can be moved, enlarged/reduced, and erased.

The operation methods for each function are as follows:

- Selection and placement of stamps
  Users select a stamp by clicking or tapping on a stamp from the stamp list, which is then automatically placed at the top-left of the illustration area. After that, users can freely reposition the stamp on their desired location on the illustration using a drag-and-drop operation.

- Moving stamps
  Placed stamps can be easily moved to any location on the illustration using drag-and-drop.
- Resizing stamps
  Users can intuitively resize stamps (enlarge or reduce) by dragging the edges of the stamps.
- Deleting stamps
  Stamps placed within the illustration area can be deleted by dragging and dropping them into the trash bin icon located at the lower-left corner of the screen.

In addition, the same word stamp can be used any number of times. Also, word stamps are classified and distributed to each part of an illustration. The system is implemented in HTML, CSS and JavaScript, and operates online. The system can be operated on a PC or tablet device, and it is possible to operate by touching the screen.

3.3. Types of word stamp

Regarding the phrases on the word stamps, we collect and analyze phrases that reflect on illustration, and create stamps with reference to the viewpoint of reflection and vocabulary that is often used. We create stamps based on the following analysis. The targets of analysis are a survey of people who draw illustrations, and illustrations posted on X and the illustration correction site 'sessa' FlatplanInc. [10]. The data from the survey of people who draw illustrations was collected from 101 men and women who draw illustrations on a regular basis. Respondents were asked "What kind of reflective sentences do you write? Please write specific sentences for each viewpoint"., and instructed to write an answer for each of the following items: hair, eyes, ears, face, clothes and accessories, hands, body, background and props, shadows, composition, other. As for the data from x and sessa, we targeted illustrations depicting people and analyzed the viewpoints and vocabulary commonly used to evaluate the illustrations. We gathered 13 illustrations from X and 10 illustrations from sessa, making a total of 23 illustrations. The data on X was gathered from posts with the hashtag '# I want you to write onto my illustration and praise me'. Regarding the reason we used not only sentences of self-reflection but also sentences expressing reflection by other people, 94% of self-reflection comprised negative reflection on parts that had not gone well, meaning it was necessary also to gather words of positive reflection. Vocabulary was extracted from the sentences and classified into the parts mentioned above, such as hair, eyes, face, etc. Furthermore, we measured the vocabulary that was most used for each part.

As a result, 1419 words were extracted. Table Table 1 is a list of the top three words for each part. The word most used in the evaluation sentences overall was 'color', which appeared 38 times. The next most common word was 'balance', appearing 34 times, then 'weird', appearing 31 times. In addition, of the total of 393 sentences extracted, phrases related to light and shadow represented 18%, while phrases related to 'color' represented 15%. From this it can be considered that, when evaluating an illustration, importance is placed on reflection related to color and light. From the frequency of the appearance of certain vocabulary, it is clear that the eyes are a part that is focused on. In addition, the fact that the word 'balance' was frequently used to evaluate the face and body parts suggests that reflection related to the position and size of parts, and the center of gravity, was focused on for the these parts.

Fig. 4 shows the stamps that were created based on the analysis of reflection phrases. A total of 41 types of stamp were prepared. There are different stamps for each part of the illustration, enabling the learner to evaluate each specific part appropriately. In addition to the categories that often appeared in the analysis of reflection sentence, i.e. hair, face, eyes, body, light and shadow, and color, there is a general category comprising stamps with words such as 'Cute' and 'I like this part', which can be used on any part of an illustration. Thus, stamps are classified into seven categories. A large number of stamps related to color and light were created to make the learner focus on reflection related to these aspects. There are eight types of stamp related to 'color' and seven types of stamp related to 'light'. These stamps make it possible for learners to carry out simple and appropriate reflection on specific aspects of their work.

All the stamps fall into one of two groups: positive stamps or negative stamps. As Reflection Stamps aims to promote positive reflection, there is a greater number of positive stamps than negative stamps, with 25 types of positive stamp and only 16 types of negative stamp.

In this study, stamps were classified as either positive or negative according to the following criteria:

- Positive stamps: Expressions explicitly containing positive evaluations or praise regarding specific features or parts of an illustration.
- Negative stamps: Expressions indicating areas requiring improvement or explicitly prompting corrections.

Stamps presented in Fig. 4, such as "Think about the lighting", "Add more hairs", "Pay attention to the flow", and "Check the perspective", were designed primarily as self-reminders to encourage improvements, and thus classified as negative stamps. However, these expressions could also be interpreted as neutral. To enhance the validity of this classification method, future research should consider introducing a new "neutral" category in addition to positive and negative categories.

4. Evaluation experiment

4.1. Experiment outline

An evaluation experiment was conducted with the aim of evaluating the usability of the proposed system and verifying its effect on the act of reflecting on illustration. The experiment collaborators each drew one picture, upon which they then reflected. The time taken for reflection, the content of reflection, the collaborators' feelings about reflection, and the System Usability Scale were taken as evaluation indices. The experiment took the form of a comparison between the 'proposed group', who used Reflection Stamps, and the 'comparison group', who used the conventional method of writing directly onto their illustrations.

The hypothesis of this experiment was as follows. It is predicted that carrying out reflection using the proposed system will reduce the time taken to reflect and diversify the viewpoints of reflection. In addition, it is predicted that negative feelings toward reflection will be reduced.

The experiment was participated in by 14 men and women in their 20 s who draw illustrations on a regular basis. The length of their experience ranged from several months to more than 10 years. Two of the participants had studied art or illustration specifically. These experiment collaborators were divided into the proposed group and the comparison group, with seven people in each group, and each group carried out reflection.

4.2. Experiment environment

To draw their illustrations, each experiment collaborator used the device and drawing application that they usually used. This was to ensure that they had no trouble operating devices or applications. Of the 14 collaborators, 10 used iPads, 3 used LCD tablets, and 1 used a pen tablet. Regarding the applications used, 7 people used CLIP STUDIO PAINT, 6 used ibisPaint, and 1 used MediBang Paint. The Surface Pro 6 was used to run the proposed system. The experiment took place in a university meeting room. There was no noise pollution or vibrations in the surroundings.

Table 1
List of top 3 most frequently occurring vocabulary for each part(()is number of appearances).

All parts	Hair	Eyes	Face	Clothing etc.	Body	Background/Props
Color(38)	Hair(18)	Light(5)	Face(15)	Clothing(8)	Balance(5)	Background(5)
Balance(34)	Flow(7)	Color(5)	Balance(8)	Wrinkles(5)	Shoulders(4)	Too(5)
Weird(31)	Color(5)	Big(5)	Part(6)	Balance(5)	Looks(4)	Perspective(4)

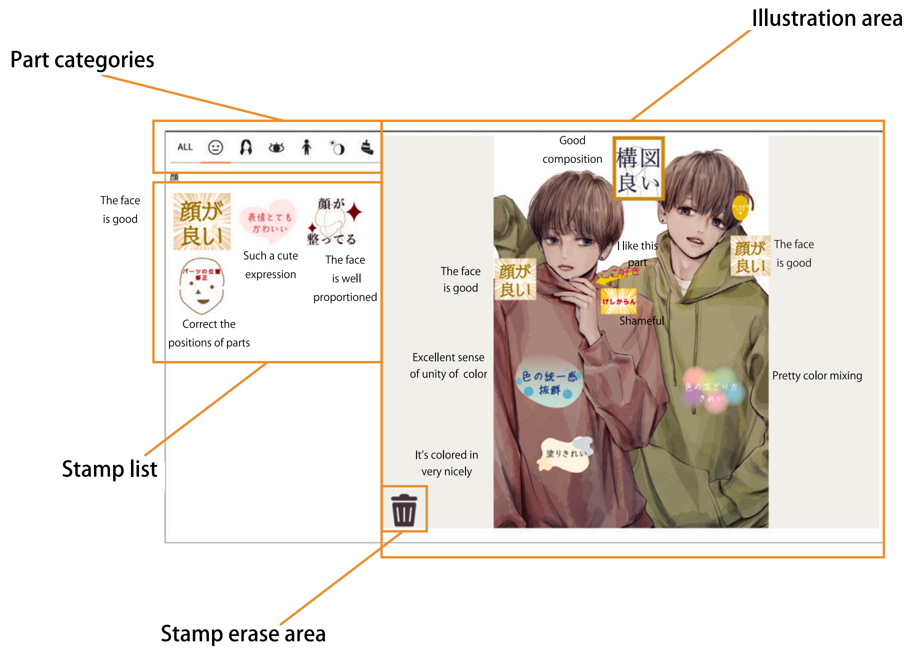

Fig. 3. Proposed system screen.

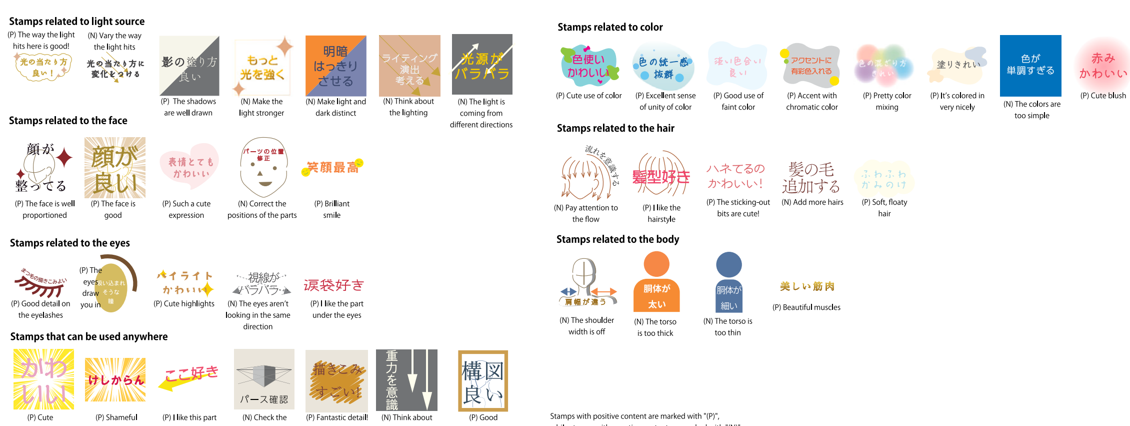

Fig. 4. List of created stamps.

4.3. Experiment procedure

First of all, the experiment collaborators were given an overview of the experiment and instructed on the conditions at the time of drawing. The conditions were as follows:

- The task is to draw a picture of a person
- The person's face must be drawn
- You can draw an original character or an existing character
- Do not draw anything that is overly grotesque or sensitive
- You do not have to complete the illustration but you should color it
- You are free to use the internet or literature to find material
- While there is a time limit, it does not matter if you exceed the time limit or finish early
- Please refrain from listening to music or watching videos etc. while drawing

Next, the collaborators were instructed to answer a questionnaire about their usual drawing methods and reflection methods. Reflection was carried out twice: once 45 min after drawing began, and once after the illustration was completed. When drawing, collaborators used the drawing process recording function of their respective drawing applications, e.g. the 'time lapse' function in CLIP STUDIO PAINT, to record their drawing processes. First, they drew for 45 min. After that, they were instructed to carry out reflection for the first time. The proposed group were instructed to use the proposed system to reflect

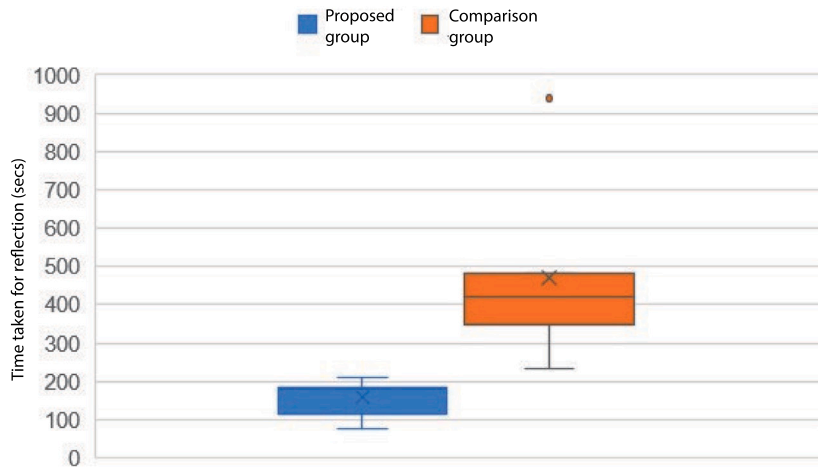

Fig. 5. Time taken for reflection.

on good points and points for improvement. At that time, they were taught how to use the proposed system. Meanwhile, the comparison group were instructed to write their reflections, as sentences, on their illustrations. As with the proposed group, they were explicitly instructed to reflect on both positive points and points for improvement. The time taken for reflection in the proposed group and comparison group, respectively, was measured, and in the case of the proposed group the screen was recorded when they operated the system. Afterwards, the collaborators were instructed to continue drawing. They were informed that they could keep drawing until their illustrations were completed. The time required for this varied between collaborators, ranging from approximately one hour to two hours. Once the illustrations were complete, reflection was carried out for the second time. This was done in the same manner as the first time. Once again, the time taken was measured in both groups. Finally, using the System Usability Scale, the proposed group were asked to rate how easy it was to use the proposed system.

The comparison group was instructed to perform reflection by handwriting sentences directly onto their illustrations. Specifically, participants recorded reflections by writing using a stylus, the same tool they usually employ when drawing illustrations. We considered it natural to adopt this approach because participants regularly use a stylus for illustration creation and reflection tasks. Furthermore, handwriting with a stylus allows participants to intuitively and easily specify the exact areas of their illustrations they wish to comment on. Therefore, we do not consider handwriting necessarily to require more time than keyboard input. However, since we cannot rule out the possibility that the difference between keyboard input and handwriting influenced the reflection time or reflective behaviors in the comparison group, additional experimental and qualitative analyses will be necessary in future research to examine the impacts of different input methods on reflection time and cognitive processes.

4.4. Experiment results

4.4.1. Time taken for reflection

The average time taken for reflection was 158 s in the proposed group and 393 s in the comparison group, which means that the proposed group took 60% less time(Fig. 5). As a result of applying a t test to the time taken for reflection in the proposed group and the comparison group, it was found that the time of the proposed group was significantly shorter ($t(11) = 6.03$, $p < .01$). Thus, it is clear that using the proposed system enables learners to easily complete the activity of reflection. Although the value was considered an outlier and excluded from the calculation of the average and the t test, it should be noted that one member of the comparison group spent more than 30 min on one reflection session, with a time of 1949 s (32 m 29 s).

4.4.2. Content of reflection

We analyzed the types of stamps and the positioning of stamps used in reflection. In the proposed group, an average of eight stamps were used in one reflection session. In the comparison group, an average of eight sentences were written in one reflection session. The content of reflections was divided into positive content and negative content, and analyzed. Fig. 6 is a graph expressing the ratio of positive to negative content in both the proposed group and the comparison group. In the proposed group, positive content represented 70% of all content. In contrast, in the comparison group, positive content represented only 23% of the total content, showing that the proposed group had a greater amount of positive reflection.

Furthermore, more stamps were used in reflection after an illustration was completed than in reflection partway through the drawing of an illustration. The collaborators presented in Fig. 7 used twice as many stamps in the post-completion reflection session than in the mid-way reflection session. In addition, there was a trend in which a greater number of positive stamps were used in post-completion reflection than in midway reflection. The number of collaborators who used more positive stamps than negative stamps was two in the midway reflection section and five in the post-completion session. This implies that learners reflect on their work more deeply once it is completed than while it is still in progress.

The collaborators' comments revealed that some were of the opinion that the variety of stamps was limited and the system required stamps that were more broadly applicable and stamps relating to the clothing and body. This feedback indicates that there is room for further improvement of the system.

Next, we explain the number of stamps used and the percentage that was used on each part of an illustration. When the first and second reflection sessions were combined, the total number of stamps used was 110. The most used stamp was 'I like this part', which expresses a part of the illustration that the learner is pleased with. Combining the first and second reflection sessions, the 7 collaborators used this stamp 18 times. The general stamps were used 36 times, representing 32% of the total. Stamps related to the face were used 23 times, representing 20% of the total. In contrast, stamps related to the eyes were used two times, and stamps related to the body were used four times. Thus, while stamps related to the face were the most widely used, stamps related to the face and body were seldom used. The collaborators in Fig. 8 are cases in which stamps were concentrated on the area around the face. In particular, stamps related to 'face' were used remarkably frequently, revealing that learners have a strong tendency to focus on this part when evaluating their work. In addition, certain collaborators used the same stamp multiple times to emphasize its meaning (Fig. 9, or enlarged stamps to express their importance(Fig. 10), showing that the use of stamps is adaptable.

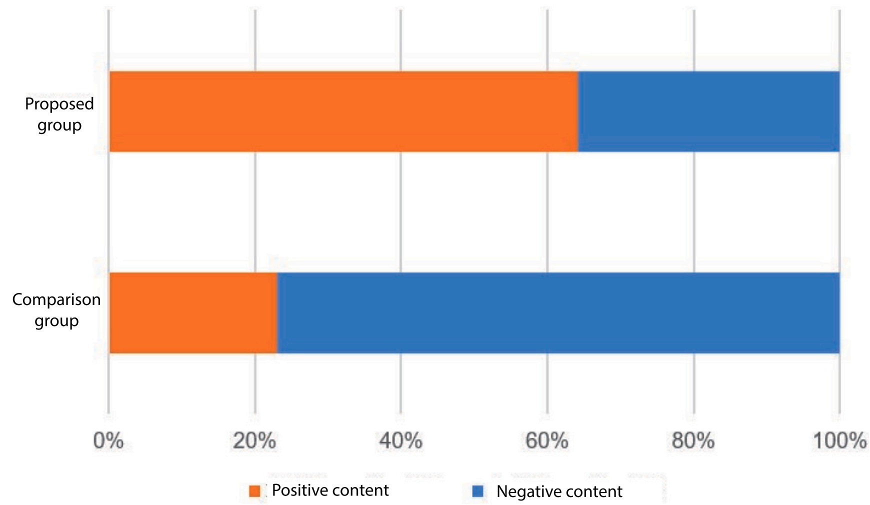

Fig. 6. Ratio of positive content and negative content in reflection content.

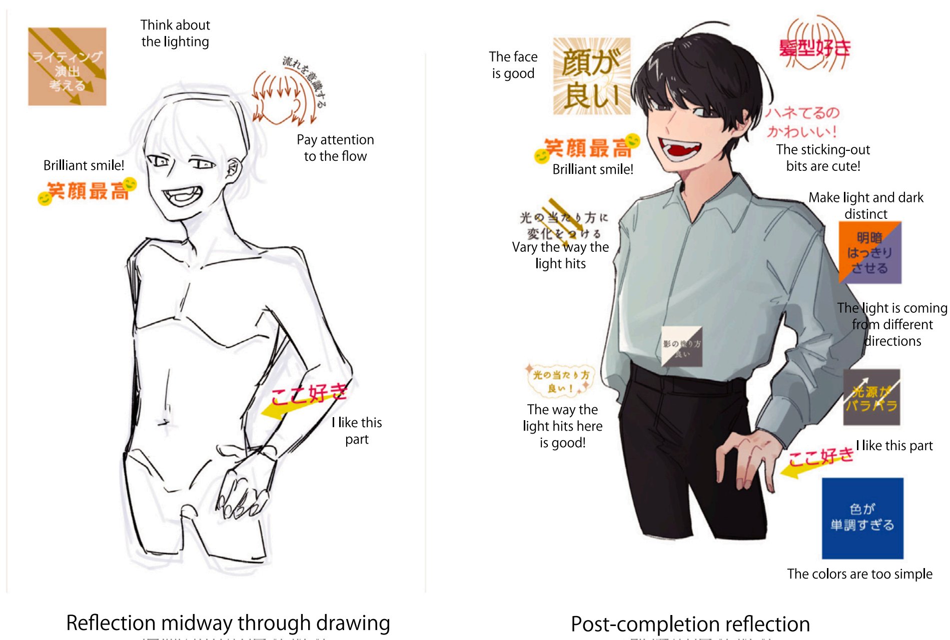

Fig. 7. Example of a difference in the number of stamps used in midway and post-completion reflection, in the proposed group.

4.4.3. Evaluation of the usability of the proposed system

The usability of the system as a whole was measured using the System Usability Scale. As a result, the score on the System Usability Scale was 79, demonstrating generally good usability. Regarding the operability of the system, the collaborators indicated several points for improvement. Firstly, they reported that the operation of placing stamps on an illustration was not intuitive and that it was especially difficult to place stamps using drag and drop. In addition, they indicated that it was difficult to enlarge and reduce the illustration, explaining that it would be preferable to perform the enlarge/reduce operation with a pinch-out movement using two fingers. We recognize that these issues present a problem regarding system usability, and that future improvements are required.

4.5. Considerations

The time taken for reflection for 60% shorter in the proposed group than in the comparison group. From this, it is clear that the proposed system has the effect of reducing the effort required for reflection and making reflection easier.

Although the System Usability Scale score was 79, which is generally good, comments from the collaborators revealed that there are problems with stamp placement and the enlargement/reduction of stamps. This means that effort is required to operate the system and thus it does not meet the requirement related to ease of use. Accordingly, it is necessary to change the stamp placement and enlargement/reduction operations to make them more intuitive. For example, rather than clicking on a stamp to select it, it would be better to be able to select and place a stamp with one operation by dragging and dropping from the stamp list onto the illustration. In addition, there was a request for a means of enlarging and reducing stamps by ‘pinching out’ with two fingers.

Regarding the content of reflection, more positive stamps were used than negative stamps, and the proposed group tended to more positive reflection than the comparison group. Therefore, we consider that the proposed system successfully encouraged positive reflection. In this

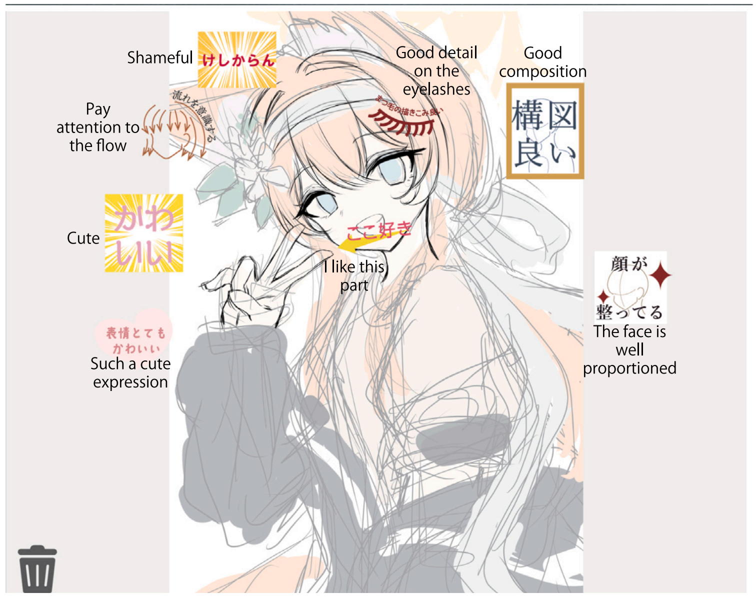

Fig. 8. Example of stamps concentrated around the Face.

Fig. 9. Actual example of using the same stamp multiple times.

study, the ratio of positive to negative stamps was intentionally set unevenly, with 25 positive stamps and 16 negative stamps. Additionally, there was an imbalance between the number of positive and negative stamps within each category. This intentional design aimed to promote positive reflection, thereby enhancing self-efficacy and maintaining learner motivation. However, this imbalance may have introduced a bias, causing participants to select positive stamps even in situations where negative feedback would have been more appropriate. Future research should examine the ratio of positive to negative stamps in greater detail and conduct additional experiments adjusting the balance, variety, and quantity of stamps. Such further research will allow for a more precise evaluation of how this design affects the content of users' reflections. Furthermore, it will be important to investigate how this imbalance in favor of positive stamp design influences users' emotions and reflective behaviors in the long term.

The experiment collaborators made comments such as "I, myself, don't try to look for positive parts, so I was really grateful that they were presented with stamps"; "The stamps are cute, which put me in a good mood"; "The stamps were cute....I was able to find good parts and points for improvement. I enjoyed drawing". The comments indicate that the collaborators in the proposed group had fun while reflecting on their work. This suggests that using the proposed system reduces the mental burden of reflection.

On the other hand, the system was found to be inferior to the comparative method in terms of accuracy and variety. The experiment collaborators commented that the variety of stamps was limited and that in some cases there were no stamps that matched their own opinions. While having a large number and variety of stamps enables more accurate and expressive reflection, it also means that more time and effort is required to search for a stamp, thus there is a trade-off between variety and simplicity. In this research, in the prior analysis described in Section 3.3, we investigated the phrases used in reflection, classified them into parts, and created stamps. Occasionally, we observed that especially when the designs drawn by the collaborators differed from those studied in the prior analysis, the stamps were not applicable. Our next task will be to create stamps for each type of design and develop a UI that enables users to select stamps that are suited to their designs.

Participants in the proposed group reflected under conditions where stamps were pre-prepared for them, whereas participants in the comparison group needed to generate the content of reflections independently. Providing such predefined reflective prompts, similar to predictive text systems Bhat et al. [16], may potentially reduce cognitive load during the reflection process. This issue warrants deeper exploration in future research.

5. Conclusion

In this research, we proposed 'Reflection Stamps', a system to reduce the effort and mental burden of the act of reflecting on one's work and encourage positive reflection. The proposed system utilizes text stamps to make the experience of reflection easier and more enjoyable. In addition, the system is designed to make it easier for learners to reflect in a self-affirming manner, by using many stamps with positive content.

In the evaluation experiment, the proposed group, who used the proposed system, took 60% less time to reflect and had 133% more positive reflections than the comparison group, who used a conventional method. Furthermore, user comments implied that the ease of using visually expressive stamps helps to reduce the mental burden of the act of reflection.

Hereafter, we intend to make the system capable of responding to a wider variety of reflections, by increasing the number of types of stamp

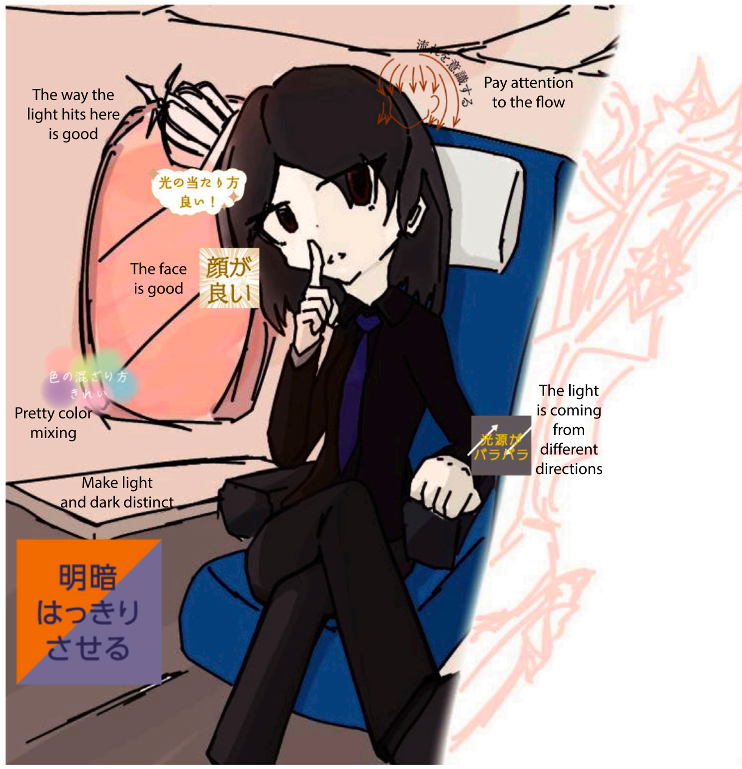

Fig. 10. Actual example of enlarging a stamp to express its importance.

and making operation more intuitive. A further task will be to improve the usefulness of the proposed system as a reflection support system, by verifying how the ratio of positive and negative reflection, and the balance of stamp content, affect users' emotions and the improvement of their technique.

Although this study conducted a short-term evaluation, it is necessary to examine the long-term effects of the proposed system on emotions, skill improvement, self-efficacy, learning motivation, and system improvements in order to accurately assess its true effectiveness. Future research should promote sustained, long-term use of the system and incorporate a long-term evaluation design that includes regular feedback and objective skill assessments.

CRediT authorship contribution statement

Seina Tajika: Writing – original draft, Validation, Software, Investigation. **Hiroki Watanabe:** Writing – review & editing. **Yoshinari Takegawa:** Writing – review & editing. **Keiji Hirata:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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